STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 8: Mechanical Properties of Solids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define stress.
2. SI unit of Young’s modulus?
3. Longitudinal strain?
4. Hooke’s law?
5. Modulus for volume change?
6. Is strain dimensionless?
7. Meaning of large Young’s modulus?
8. Longitudinal stress?

Section B 2-mark questions - 6 x 2 = 12 marks

9. N on 2 x 10- m^2: stress?
10. m wire extends 1 mm: strain?
11. Stress=2 x 10 Pa, strain=10-: Y?
12. cm cube compresses by .1 cm: strain magnitude?
13. L=1 m,A=2 x 10- m^2,Y=2 x 10,F=40 N: extension?
14. Why use steel reinforcement in concrete?

Section C 3-mark questions - 4 x 3 = 12 marks

15. Wire L=2m,A=10-,Y=2 x 10,F=100N: extension?
16. Stress 4 x 10 Pa at strain 2 x 10-. Estimate modulus.
17. B=1.5 x 10 Pa, volume strain=-2 x 10-. Pressure increase?
18. Why does doubling wire length double its extension if F,A,Y fixed?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Distinguish proportional and elastic limits.
20. Derive extension and calculate for F=50 N,L=1.5m,A=10-,Y=2 x 10.